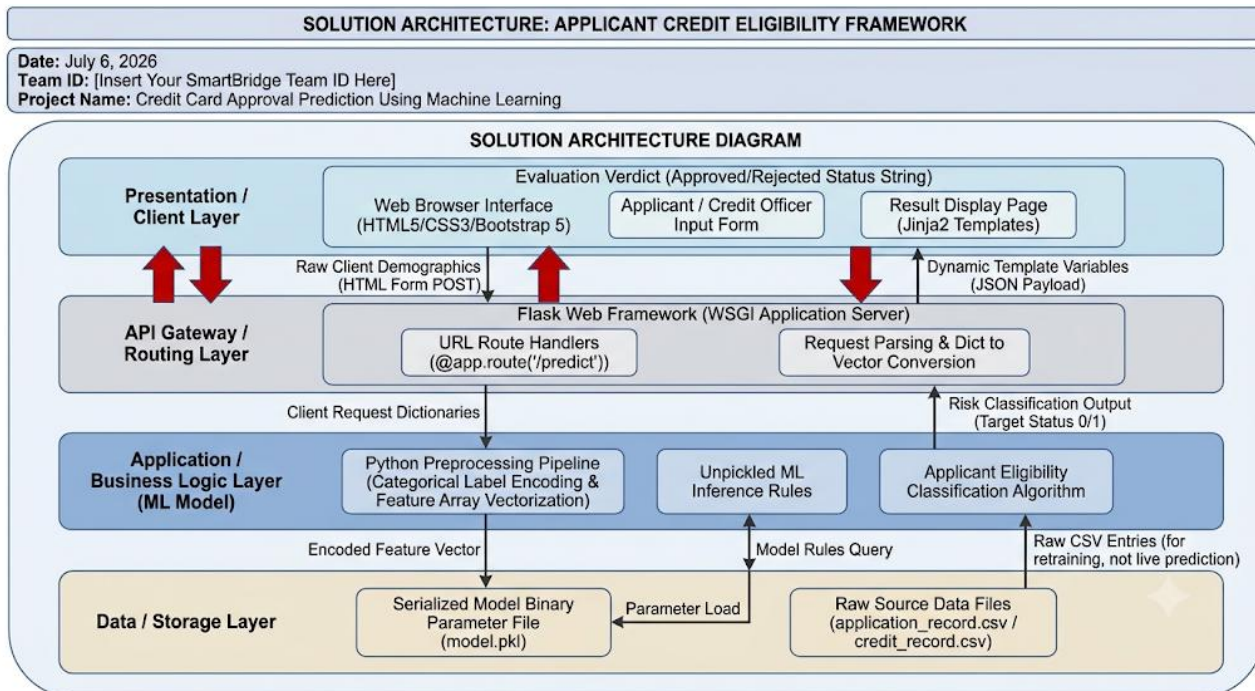


Solution Architecture

Date	29 june 2026
Team ID	xxxxxxx
Project Name	Credit card approval prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	<i>Renders an intuitive, client-facing web application interface containing input fields and selection dropdowns to collect demographic properties from the user. It intercepts client submission events and dynamically displays the model's final risk evaluation output string.</i>	HTML5, CSS3,
[API Gateway]	<i>Acts as the operational backbone for incoming traffic. It intercepts HTTP POST payload packets transmitted from the frontend form submission, parses the client request dictionaries, passes them to backend execution services, and routes response outputs back to the client browser.</i>	Flask Web Framework (WSGI Application Server)
Core Logic Service (Machine Learning)	<i>Extracts form strings, map-transforms categorical texts into optimized numerical arrays, loads the serialized baseline metrics, and executes predictive math matrix arrays to evaluate if an applicant profile represents an acceptable credit risk..</i>	Python 3, Scikit-Learn, NumPy
[Data / Storage Layer]	<i>Houses the underlying raw datasets used during structural algorithm benchmarking and training cycles. It permanently retains the final decision-tree boundary rulesets inside a lightweight serialized file asset for instantaneous real-time prediction lookup.</i>	Pandas DataFrames, Pickle Serializer Binary Asset (model.pkl)